CCIT Report No. 375

Large deviations with diminishing rates

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ABSTRACT

The theory of large deviations for jump Markov processes has been generally proved only when jump rates are bounded below, away from zero [2, 5, 7]. Yet various applications of interest do not satisfy this condition. We describe several classes of models where jump rates diminish to zero in a Lipschitz continuous way. Under appropriate conditions, we prove that the sample path large deviations principle continues to hold. Under our conditions, the rate function remains an integral over a local rate function, which retains its standard representation